La Scuola d'Italia / Dr. Huson / IB Math 12
14 October 2026

Nickname:

1.11 Quiz: Statistics

1. [Maximum mark: 7]

   The table shows the waiting time, t minutes, of 50 patients at a clinic.

Time, t	$0 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 30$	$30 \leq t < 40$	$40 \leq t < 50$
Number of patients	4	12	18	11	5

   (a) State whether the data are discrete or continuous. [1]

   (b) Write down the modal class. [1]

   (c) Find an estimate of the mean waiting time. [2]

   (d) Find an estimate of the standard deviation. [1]

   (e) The 50 patients are a stratified sample, by age group, of the 600 patients registered at the clinic. There are 192 registered patients over the age of 65. Find the number of patients over the age of 65 in the sample. [2]

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2. [Maximum mark: 9]

Fifteen students recorded the number of push-ups they could do in one minute.

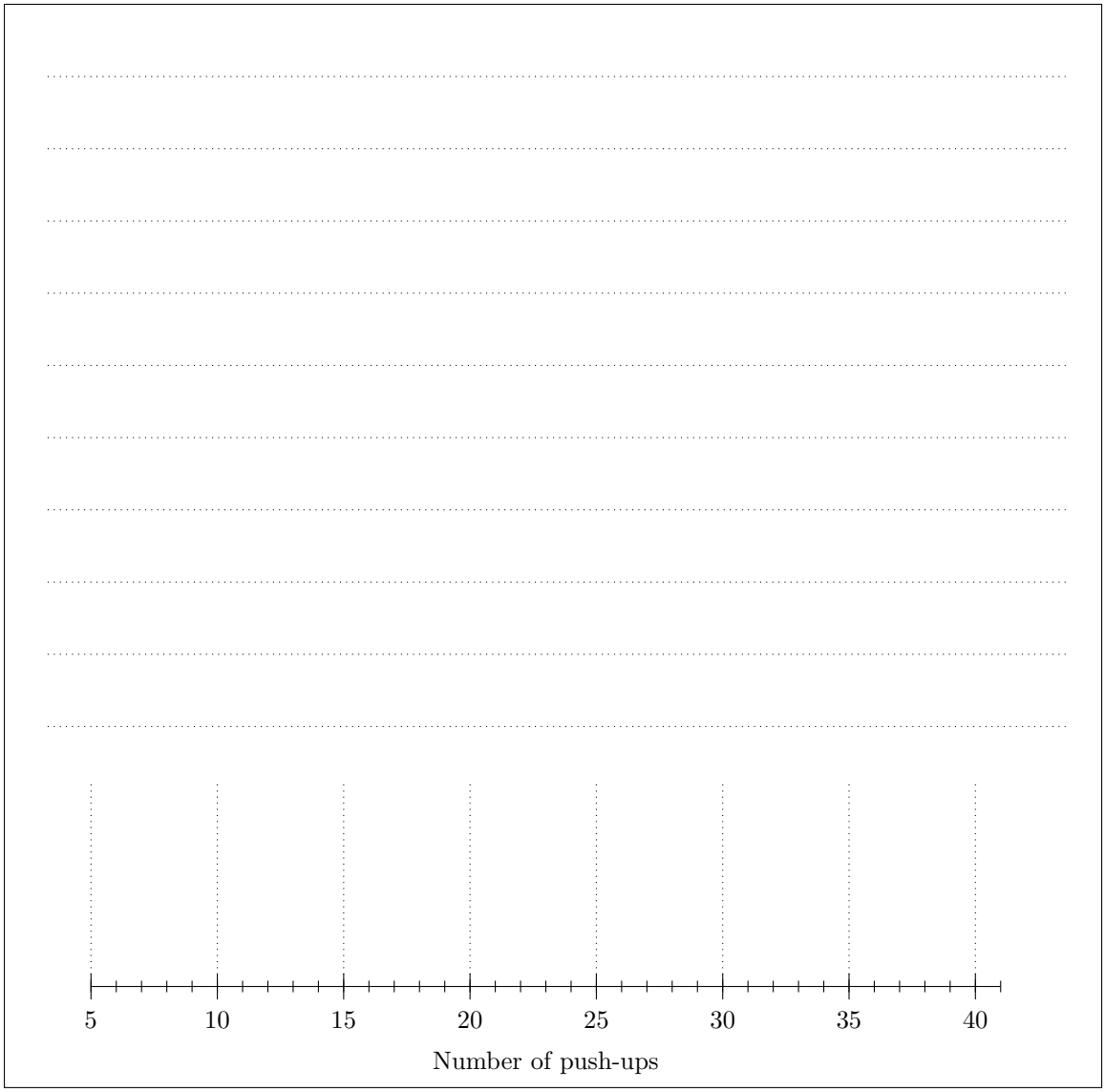
27, 31, 22, 35, 28, 25, 33, 8, 29, 24, 36, 26, 30, 28, 32

(a) Write down the median, the lower quartile Q_1 and the upper quartile Q_3. [2]

(b) Find the interquartile range. [1]

(c) Show that 8 is an outlier. [3]

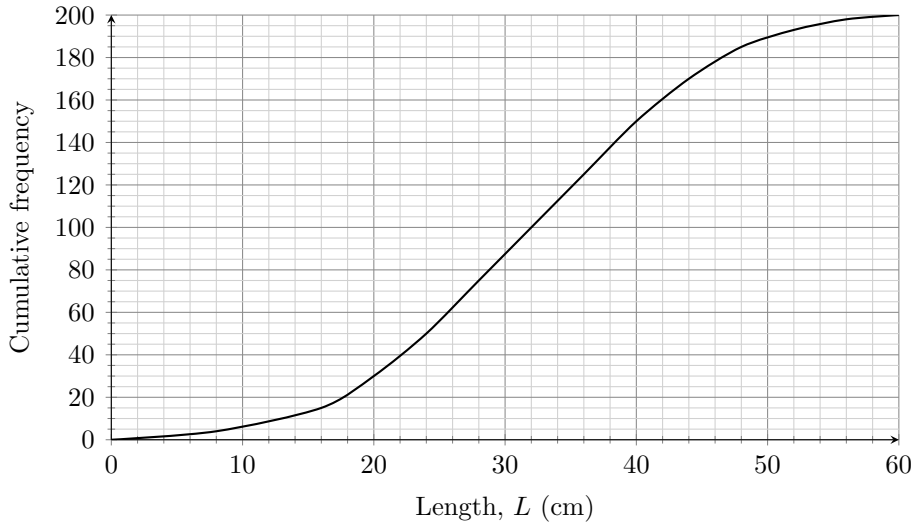
(d) On the scale at the bottom of the box, draw a box-and-whisker plot of the data. Mark the outlier with a cross. [3]



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3. [Maximum mark: 9]

The cumulative frequency graph shows the lengths, L cm, of 200 fish caught in a river.



(a) Write down the median length. [1]

(b) Find the interquartile range. [2]

(c) Find the number of fish longer than 44 cm. [2]

(d) The shortest 15% of the fish are returned to the river. Find the greatest length of a fish that is returned. [2]

(e) The lengths are grouped in the table below. Find the values of p and q. [2]

Length, L (cm)	$0 < L \leq 20$	$20 < L \leq 28$	$28 < L \leq 44$	$44 < L \leq 60$
Number of fish	30	p	q	30

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4. [Maximum mark: 8]

The table shows the age, x years, and the price, y thousand dollars, of eight used cars of the same model.

Age, x (years)	1	2	3	5	6	8	9	10
Price, y (thousand dollars)	27	27	22	21	16	15	11	12

(a) Find Pearson's product-moment correlation coefficient, r. [1]

(b) Characterize the correlation (positive or negative; strong, moderate, weak, or none). [2]

(c) Write down the regression line of y on x, in the form $y = ax + b$. [2]

(d) Interpret the meaning of the value of a in this context. [1]

(e) Use your regression line to estimate the price of a car that is 7 years old. [2]